

Predicting the Market Potential of Extended Range Electric Vehicles in the United States

Alex Ponnraj

Abstract

This paper evaluates the prospective United States market potential of EREVs (Extended Range Electric Vehicles) using historical monthly vehicle sales, vehicle specifications, and LightGBM regression models. The analysis is motivated by the lack of direct EREV sales history in the United States and by Scout Motors planned Harvester range extender, which is described by Scout as a gas powered generator that recharges the high voltage battery and is expected to raise total range to more than 500 miles. The central modeling challenge is that a hypothetical vehicle launch has no observed sales history, so the model must rely on static vehicle attributes, launch timing, powertrain identity, pricing, performance, and segment signals. To address this challenge, two no lag launch models were evaluated using different target formulations: raw monthly sales and log transformed monthly sales. The raw sales model achieved stronger performance in actual unit sales terms, with a mean absolute error of 380 vehicles, RMSE of 794 vehicles, and R squared of 0.82. The log sales model achieved a log RMSE of 1.27 and log scale R squared of 0.84, but produced higher error when converted back to unit sales, with a mean absolute error of 472 vehicles and RMSE of 1,105 vehicles. Because the objective is to estimate monthly demand for a hypothetical Scout launch, the raw sales model was selected as the primary forecasting approach. Feature importance results suggest that early launch performance is most strongly associated with launch timing, price per horsepower, seasonality, brand, relative segment price, power to weight ratio, and efficiency. These findings indicate that, even without prior sales history, vehicle attributes and market positioning can provide useful signals for estimating early adoption potential.

Introduction and Problem Statement

The automotive industry is undergoing a rapid transition driven by electrification, emissions regulation, changing consumer expectations, and the pressure to reduce dependence on conventional internal combustion vehicles. Battery electric vehicles have gained visibility and market share, yet they still face barriers that matter to many American customers. Charging availability, charging speed, towing range, cold weather range loss, and long distance travel anxiety can make a purely electric vehicle feel risky for households that expect one vehicle to satisfy both daily commuting and occasional long range travel.

An Extended Range Electric Vehicle addresses this gap by operating primarily as an electric vehicle while carrying a gasoline powered generator to recharge the battery when needed. In this configuration, the engine does not behave like a traditional combustion drivetrain. The electric motors remain responsible for propulsion, while the generator extends operating range after the battery reaches a lower state of charge. This architecture creates a bridge between the convenience of liquid fuel and the driving feel of an electric vehicle. It also creates a market forecasting problem because EREVs are not yet represented with enough historical United States sales observations to train a model directly on EREV adoption patterns.

Scout Motors makes this problem especially relevant. Scout has presented the Traveler SUV and Terra pickup as rugged electric vehicles with an available Harvester range extender. Scout describes the Harvester system as a generator packaged into the vehicle platform that recharges the high voltage battery and supports an expected total range above 500 miles. Public reporting also indicates that Scout adopted the range extender option in response to United States driver concerns about charging and long distance

use. For this project, the Scout Traveler EREV is treated as a hypothetical new entrant whose demand must be inferred from related historical vehicles rather than from direct EREV sales history.

The purpose of this project is to estimate the potential market performance of a Scout EREV by training machine learning models on historical vehicle sales and specifications. The analysis uses a custom sales parser, lifecycle filtering, feature engineering, and 2 LightGBM models.

Data Collection

The data foundation for this project required transforming raw sales files into a model ready dataset. The source sales data came from MarkLines style tables containing United States monthly sales by maker, model, powertrain, and year. The raw structure was not already suitable for modeling because it was hierarchical, wide, and inconsistent. Vehicle rows were embedded within tables that also contained maker groups, model groups, powertrain labels, monthly columns, and aggregate rows. A custom parser was therefore created to extract model level and powertrain level observations from raw HTML and RTF files and convert them into a consistent master sales dataset.

The parser flattened the original table structure into a long analytical format. Each observation was standardized around maker or brand, model, powertrain, date, year, month, and monthly sales. This was necessary because the original files represented months as separate columns. The parser converted the monthly columns into a single date field in year month format and one sales value per row. This long format made it possible to merge sales records with vehicle specification records by maker, model, powertrain, and year. It also allowed launch timing and lifecycle features to be calculated month by month rather than only year by year.

Merging Vehicle Specifications

The sales records alone were not sufficient for demand prediction, so they were merged with a master vehicle specification dataset. The specification data included acceleration, top speed, fuel economy or electric equivalent efficiency, range, horsepower, torque, weight, payload, cargo volume, body style, seating capacity, launch price, and a luxury tier score. Because vehicle specifications can change across years, the merge was designed around the sale year. When the exact year specification existed for a given maker, model, and powertrain, that year was used. When the exact year was missing, the nearest prior year specification was carried forward. This approach follows the assumption that if a model was sold in a year but a new specification row was unavailable, the latest known prior specification was the most reasonable proxy.

Comparing Efficiency Across EVs vs ICEs

Efficiency was compared across EVs, hybrids, and traditional gasoline vehicles by converting fuel economy into a shared energy intensity metric, `energy_kwh_per_100mi`. This allowed all powertrains to be evaluated on the same scale, where lower values represent better efficiency. For EVs, MPGe was converted using the EPA gasoline-equivalent value of 33.7 kWh per gallon equivalent:

$$\text{energy_kwh_per_100mi} = 3370 / \text{mpge}$$

For gasoline and hybrid vehicles, MPG was converted into an equivalent energy-use measure using the energy content of gasoline. The applied conversion was:

$$\text{energy_kwh_per_100mi} = (235.214 / \text{mpg}) * 8.9$$

where 235.214 / mpg converts MPG into liters per 100 kilometers, and 8.9 kWh/L approximates gasoline's energy content. This transformation provided a consistent efficiency feature across vehicle types, allowing the model to compare electric and non-electric vehicles using a single standardized measure rather than separate MPG and MPGe columns.

Vehicle Lifecycle Cleaning

Lifecycle cleaning was a critical part of the data preparation process. The raw sales panel naturally included zero sales for vehicles before launch and after discontinuation because the wide table contained columns for every month in the collected period. If those rows remained in the training data, the model could learn false demand patterns. For example, a vehicle that did not exist in 2016 could appear as a long sequence of zero sales until its real launch year. Those zero observations do not represent poor market demand. They represent non availability. The cleaning pipeline therefore referred to the launch timing from a launch map indicating the release year and month for specific vehicle maker, model, powertrain combinations for the US market, then removed pre launch rows, including months inside the first calendar year before the first observed sale.

A different approach was used for discontinued vehicles. Long sequences of zero sales after a vehicle exited the market can appear in the raw panel. These observations can make the model think demand collapsed when the more accurate interpretation is that the vehicle was no longer offered. The pipeline identified the last positive sales month and filtered rows after that point when appropriate. This created a more realistic active sales window for each vehicle. The resulting dataset better reflects genuine market behavior during periods when the vehicle could actually be purchased.

Feature Engineering

Feature engineering was used to make the model more sensitive to relationships that are not obvious from the raw variables. The project converted categorical fields such as maker or brand, powertrain, simplified powertrain, and luxury tier into categorical types for LightGBM. Sales were transformed using the natural log of one plus sales. This helped reduce the influence of very high volume vehicles while preserving information from lower volume models. Price was also log transformed to reduce skew in the pricing distribution.

Performance Ratios

Several ratio features were added to capture relationships between vehicle attributes rather than relying only on raw values. The power_to_weight feature measures horsepower relative to vehicle weight, which provides a better representation of performance potential than horsepower alone. The price_per_hp feature measures how much a buyer pays for each unit of horsepower. This captures the relationship between cost and performance, helping distinguish vehicles that offer strong performance value from vehicles that are priced at a premium relative to their output.

Relative Segment Pricing

The price_vs_segment feature was created to compare each vehicle's price against the average price of vehicles within the same luxury tier. This is important because price alone does not fully describe whether a vehicle is expensive or affordable in its competitive context. For example, a \$55,000 vehicle may be considered expensive in an economy segment but relatively affordable in a luxury segment. By dividing each vehicle's price by the average price of its luxury category, the model can better identify whether a vehicle is positioned below, near, or above the expected price level for its class.

Seasonality

Seasonality was represented using sine and cosine transformations of month. This is better than treating month as a simple integer because automotive sales are cyclical. December and January are adjacent in the calendar, but an integer representation would make them appear far apart. The sine and cosine features preserve this circular structure. The project also created months since launch, a lifecycle variable that describes how long a vehicle has been available. This feature is especially important for comparing mature vehicles with newly introduced models, because launch ramp up and later lifecycle decay affect sales behavior

Data Preview

Maker	Model	Power Train	Date	Sales	Year	Month	Accel_sec	TopSpeed_mph	energy_kwh_per_100mi
ACURA	MDX	ICE	2017-01	6548	2017	1	6.5	130	95.1
ACURA	MDX	ICE	2017-02	7082	2017	2	6.5	130	95.1
ACURA	MDX	ICE	2017-03	8282	2017	3	6.5	130	95.1
ACURA	MDX	HV	2017-04	114	2017	4	6.4	130	77.5
ACURA	MDX	ICE	2017-04	7990	2017	4	6.5	130	95.1
ACURA	MDX	HV	2017-05	296	2017	5	6.4	130	77.5

Figure 1. Abbreviated Data table used for model training. Sales data unpivoted into a long format and joined with vehicle specs on Maker, Model, PowerTrain, and Year

Modeling Strategy

The revised modeling strategy began with two LightGBM regression models designed to predict first year vehicle launch performance. The first model used raw monthly `Sales` as the target variable, while the second model used log transformed monthly sales, defined as `Sales\_log = log1p(Sales)`. Comparing these two target definitions allowed the project to evaluate whether the model performed better when learning sales directly or when reducing the influence of extremely high volume vehicles through a log transformation.

Both models were trained only on observations within the first twelve months after vehicle launch, where `months\_since\_launch` was between 0 and 12. This filtering step shifted the modeling problem away from mature lifecycle sales patterns and toward launch period behavior. This was important because the central goal of the study was to estimate the potential sales performance of a hypothetical Scout EREV, which would not have prior sales history at the time of prediction. By training on launch period observations, the models learned from vehicles during the same lifecycle stage that the Scout EREV forecast attempts to represent.

LightGBM was selected because it can model nonlinear relationships, handle both numeric and categorical variables, and capture interactions between features such as maker or brand, powertrain, price, range, efficiency, performance, body style, and seasonal timing. The models used pre launch or launch available features rather than lagged sales variables. Although lag based features can improve predictive accuracy for existing vehicles, they require prior monthly sales observations and therefore are not available for a new vehicle before launch. Excluding lag columns made the final modeling framework more appropriate for forecasting a new nameplate.

The raw sales model produced predictions directly in monthly unit sales, while the log sales model produced predictions on the transformed sales scale and then converted them back to unit sales using `Sales = expm1(Sales\_log)`. This comparison helped evaluate two different modeling assumptions: whether launch sales should be modeled directly in their original scale, or whether the target should be compressed to make the model less dominated by unusually high volume launches. Both approaches were retained for comparison, with final interpretation based on their predictive performance and suitability for the Scout EREV forecasting task.

The dataset was split into training and testing subsets using a 70/30 random train test split. The training set contained 1,812 observations and 24 input features, while the test set contained 777 observations and the same 24 features. The test_size = 0.30 setting reserved 30 percent of the launch period observations for out of sample evaluation, while the remaining 70 percent were used to train the LightGBM model. A fixed random_state = 42 was used so that the same rows would be assigned to the training and testing sets each time the notebook was run, improving reproducibility. The rows were also shuffled before splitting, so the validation set was randomly sampled across vehicles and launch months rather than separated chronologically.

Modeling Results

The two launch year LightGBM models showed strong predictive performance, suggesting that first year sales behavior can be meaningfully modeled using vehicle specifications, pricing, brand, seasonality, and launch timing. The raw sales model achieved an MAE of approximately 372 vehicles, an RMSE of approximately 795 vehicles, and an R^2 of 0.822. This indicates that the model explained a large share of variation in monthly launch period sales while maintaining relatively low average prediction error on the original sales scale. Its most important features were `months\_since\_launch`, `price\_per\_hp`, `month\_cos`, `Maker/Brand`, `month\_sin`, and `price\_vs\_segment`, suggesting that launch timing, seasonal demand patterns, brand effects, and value relative to performance or segment pricing were major drivers of predicted sales. The log sales model also performed well, achieving an RMSE of 1.271 on the log scale and an R^2 of 0.836. When converted back to unit sales, it produced an MAE of approximately 472 vehicles and an RMSE of approximately 1,105 vehicles.

Model	Target Variable	MAE Sales	RMSE Sales	R ²	RMSE Log
Raw Sales Model	Sales	372.60	795.92	0.822	N/A
Log Sales Model	Sales_log = log1p(Sales)	471.88	1,104.76	0.836	1.271

Figure 2. Two LightGBM models were compared using different target variables. The Raw sales model was later chosen for sales predictions of Scout EREV vehicles.

Although the log model had a slightly higher R² on the transformed target, the raw sales model produced lower error when evaluated in actual monthly vehicle sales. Since the next stage of the study requires forecasting expected unit sales for the Scout Motors vehicles, the raw sales model was selected as the primary prediction model. This choice made the results more directly interpretable and better aligned with the study's goal of estimating monthly and annual sales volumes for hypothetical Scout EREV launches.

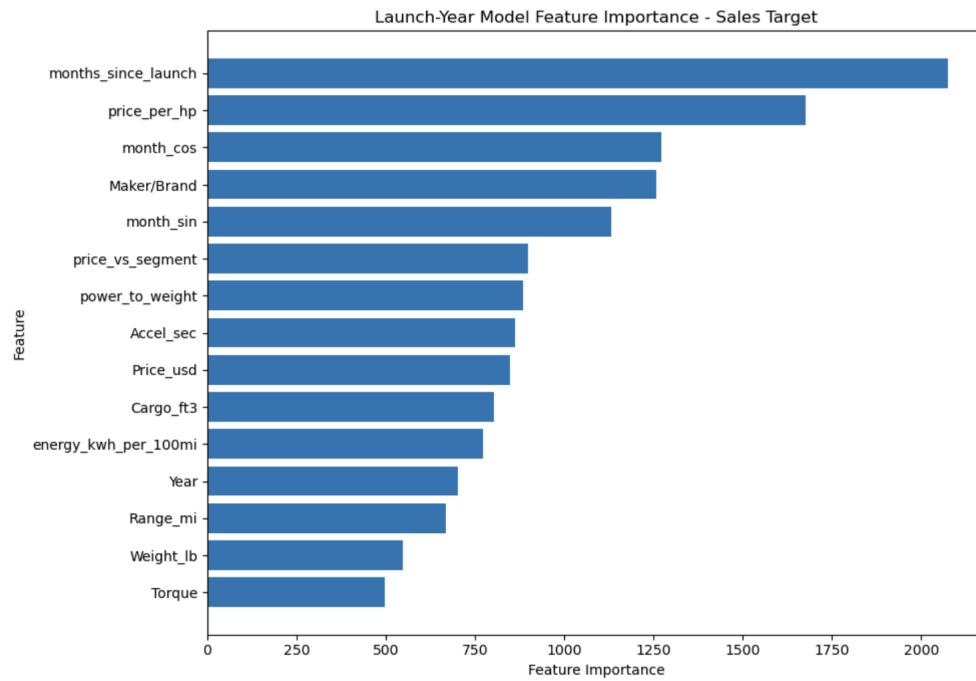


Figure 3. The Feature importance of the Sales target model.

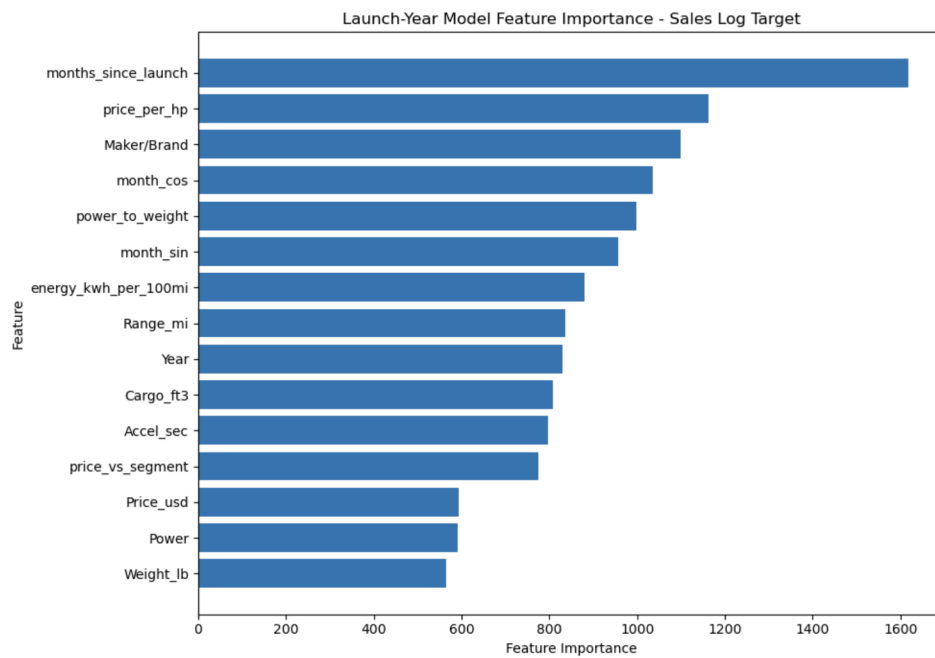


Figure 4. Feature importance of the Log Sales target model.

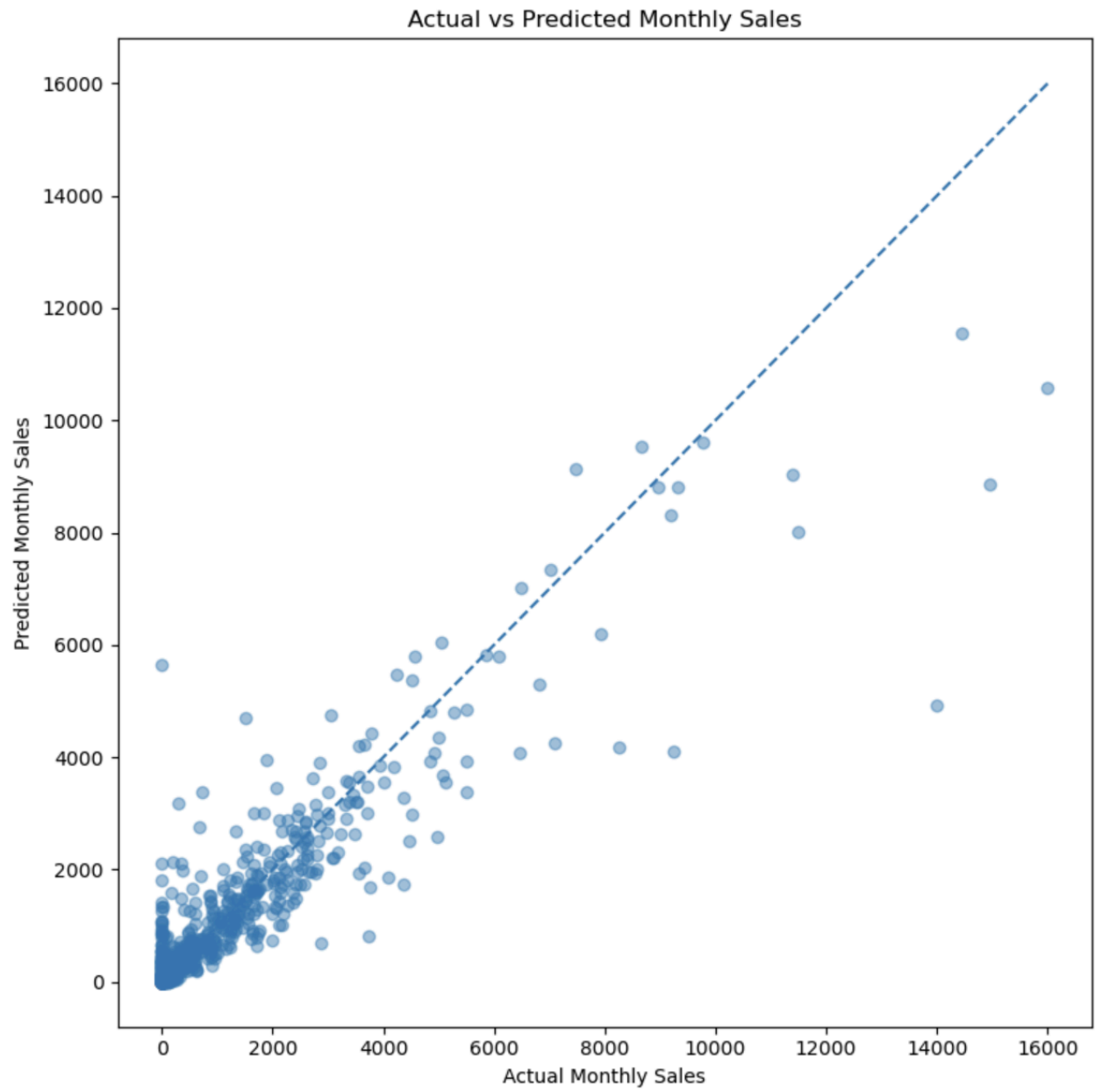


Figure 5. Actual vs Predicted Monthly sales of the Sales target model.

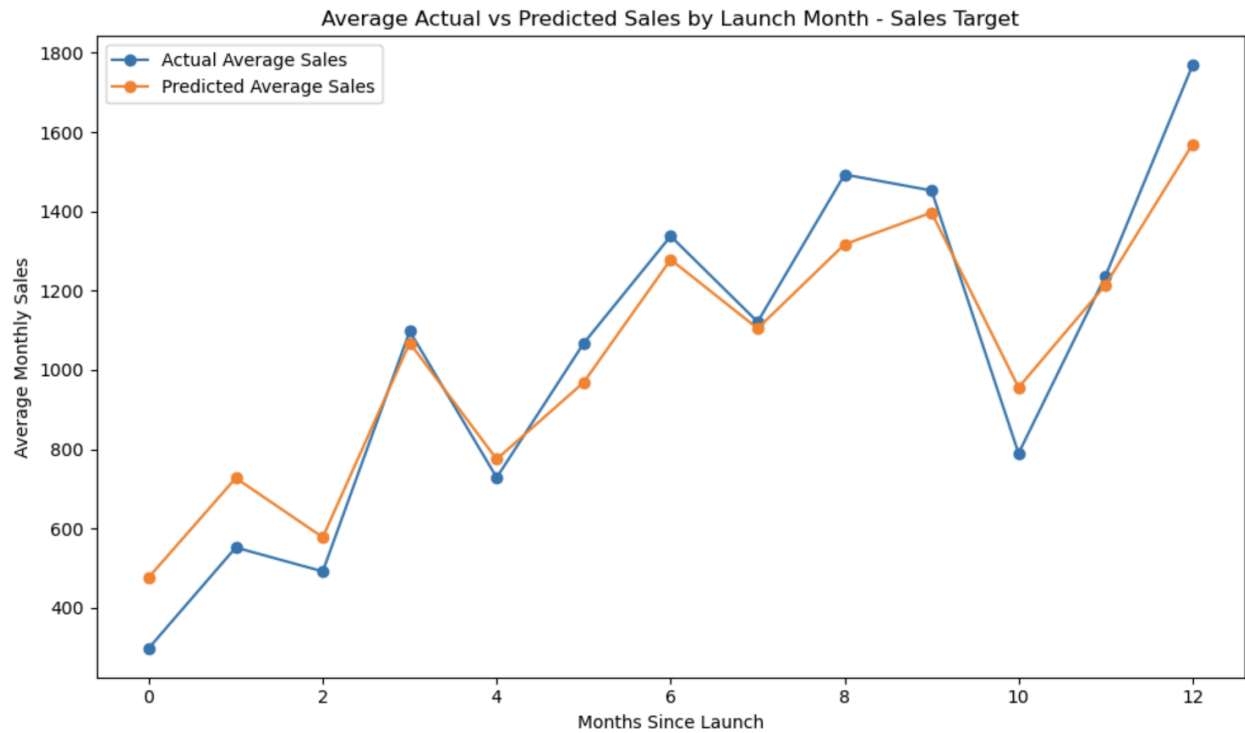


Figure 6. Actual Average vs Predicted Average Sales by Month of the Sales target model.

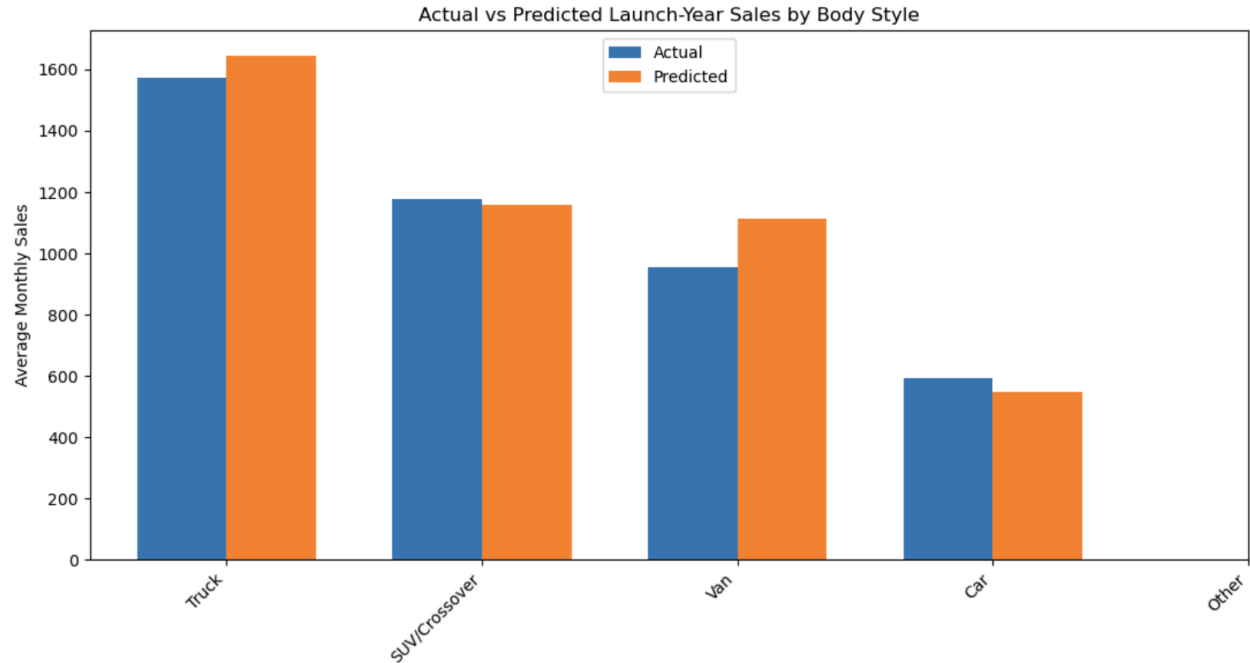


Figure 7. Actual vs Predicted Sales by Body Style of the Sales target model.

EREV Sales Forecasting

After evaluating the launch year sales model, the final stage of the analysis applied the trained model to a hypothetical Scout Motors EREV vehicle. Because this vehicle does not yet have observed sales history, the prediction process required constructing a synthetic vehicle profile using the same feature structure as the training dataset. The hypothetical vehicle row included static product attributes such as price, range, power, torque, weight, payload, cargo volume, seating capacity, body style, luxury classification, and powertrain type. Additional engineered features were also calculated to match the model inputs, including log price, power to weight ratio, price per horsepower, price relative to segment, and cyclical month indicators. To simulate the first year of launch, the same base vehicle profile was repeated across twelve monthly observations, with `months\_since\_launch` increasing from 0 to 11 and the seasonal month sine and cosine features changing month by month. Since the chosen model was trained on actual monthly sales rather than log transformed sales, the resulting predictions were interpreted directly as monthly unit sales.

Forecasting Scout Traveler SUV

```
Scout_Traveler = {
    "Maker/Brand": "VW",
    "PowerTrain": "EREV",
    "PowerTrain_simple": "Hybrid",
    "BodyStyle": "SUV/Crossover",

    "Year": 2027, #Estimated Release
    "Accel_sec": 3.5,
    "TopSpeed_mph": 155,
    "energy_kwh_per_100mi": 45,
    "Range_mi": 500,

    "Power_hp": 500, #Estimated Specs
    "Torque_ftlb": 900,
    "Weight_lb": 6200,
    "Payload_lb": 1400,
    "Cargo_ft3": 85,

    "Seats": 5,
    "Price_usd": 60000,
    "Luxury": 2,

    "log_price": np.log1p(60000),
    "power_to_weight": 500 / 6200,
    "price_per_hp": 60000 / 500,

    "price_vs_segment": 1.10}
```

One important adjustment was the use of Volkswagen as the predictive brand proxy rather than entering Scout as an entirely new brand. Brand was one of the strongest predictors in the model, but Scout does not have historical sales data in the dataset. If Scout were entered directly, the model would treat it as an unseen categorical value, which could weaken the reliability of the prediction. Volkswagen was selected as the proxy because it is the parent company of Scout Motors and therefore provides a more grounded representation of expected brand scale, distribution capacity, and market presence than an unknown category. This choice does not imply that Scout will perform identically to Volkswagen, but it gives the model a practical reference point for estimating launch performance. The final predictions were then compared against benchmark vehicles, including similar electric and electrified SUV or crossover models, to evaluate whether the hypothetical Scout EREV forecast appeared reasonable within the broader market context.

Benchmark Vehicles

Benchmark vehicles were selected based on similarity to the hypothetical Scout Traveler EREV in body style, powertrain, and price range. Vehicles were also required to have sufficient launch data available, with the initial twelve months of sales used where possible. To create a fairer comparison, certain models were excluded when their sales were likely influenced by an already established nameplate rather than demand for a truly new vehicle. For example, vehicles such as the Acura MDX Hybrid and Jeep Wrangler Hybrid represent newer powertrain combinations, but the underlying model names already had strong market recognition before the electrified version launched. This historic nameplate strength can inflate early sales relative to a newly introduced vehicle. As a result, some of these models were removed from the benchmark set so that the comparison better reflected launch performance for newer electrified SUVs and crossovers.

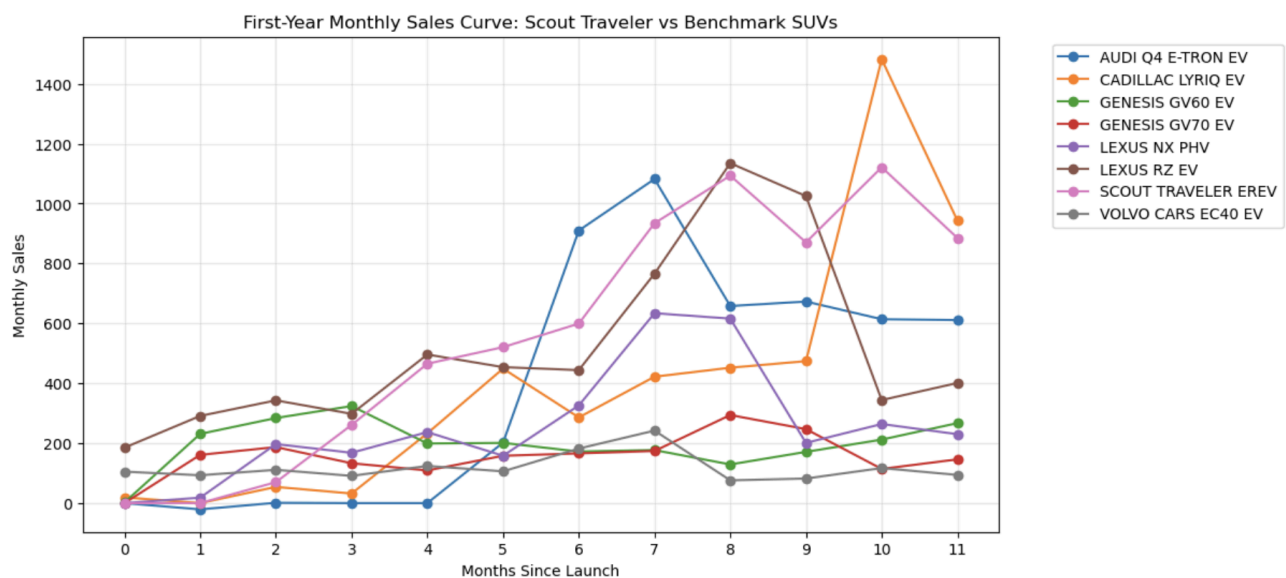


Figure 8. First Year Monthly Sales Curve of Scout Traveler vs other new Make and Model SUV releases.

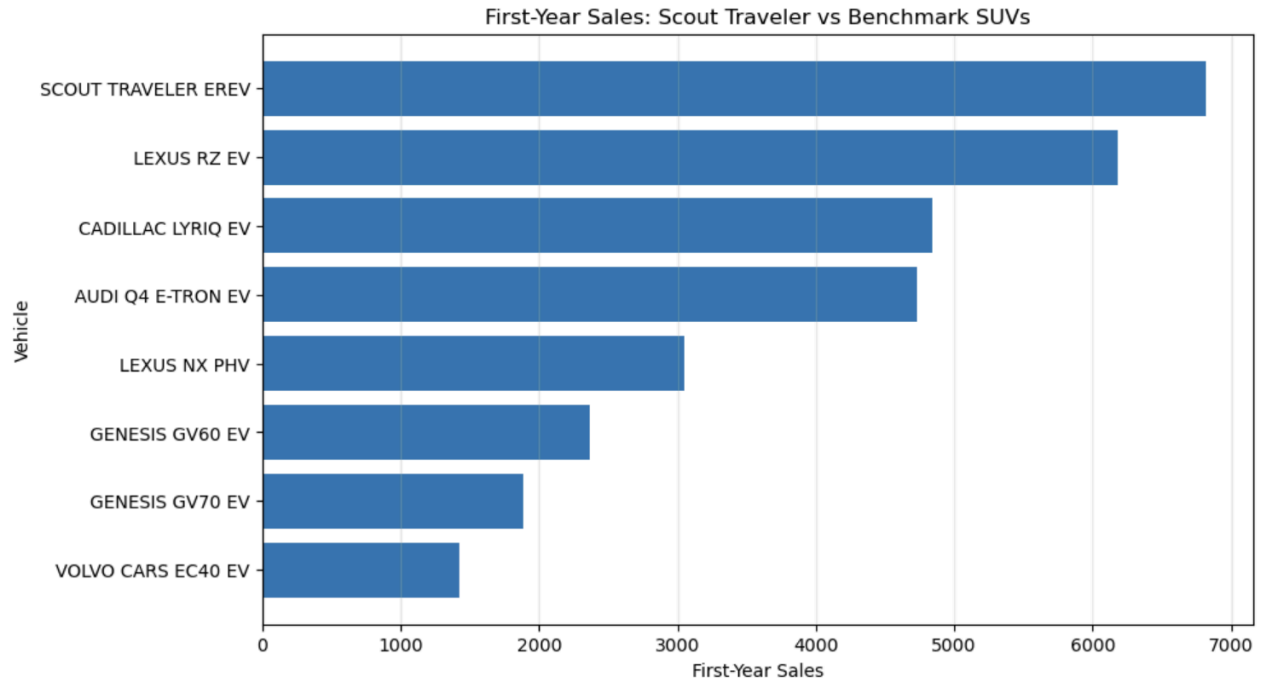


Figure 9. First Year Sales Total of Scout Traveler vs other new Make and Model SUV releases.

Forecast Against Benchmark Vehicles

Based on the benchmark comparison, the hypothetical Scout Traveler EREV prediction appears relatively strong within the selected electrified SUV and crossover launch set. The model estimates approximately 7,815 first year sales, with an average of about 651 units per month and a peak monthly prediction of roughly 1,137 units. This places the Scout Traveler above the selected similar SUV benchmarks in the file, including the Audi Q5 PHV at 6,965 units, Lexus RZ EV at 6,182 units, and the Cadillac Lyriq EV at 4,846 units. The prediction curve also shows a gradual ramp after the first two months, suggesting that the model expects limited early launch volume followed by stronger sales momentum later in the first year.

Overall, the result suggests that the Scout Traveler EREV would perform toward the upper end of the selected benchmark group rather than as a low volume niche entrant. This outcome is likely influenced by the vehicle's strong range, high power output, SUV body style, competitive estimated price, and the use of Volkswagen as the brand proxy, which gives the model an established manufacturer signal rather than treating Scout as an unknown category. However, the forecast should still be interpreted cautiously because the model is estimating demand for a new brand and a relatively uncommon powertrain configuration in the U.S. market. The benchmark comparison is therefore most useful as a directional validation, showing that the predicted first year volume is plausible when compared with similarly priced electrified SUVs and crossovers, but not necessarily a precise sales forecast.

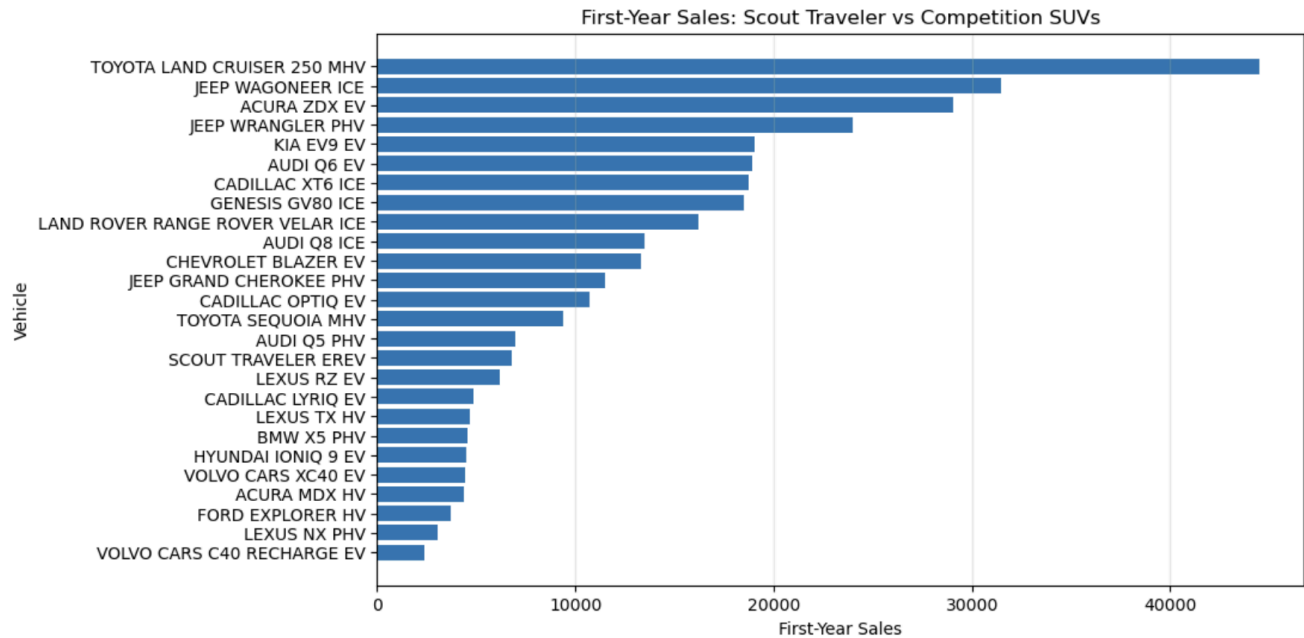


Figure 10. First Year Monthly Sales Curve of Scout Traveler vs similar segment SUVs.

Forecast Against Segment

The Scout Traveler EREV forecast places the vehicle in the lower middle of the similar SUV benchmark group, with predicted first year sales of approximately 6,821 units, an average of 568 units per month, and a peak monthly sales estimate of about 1,121 units. This places the Traveler below stronger launch performers such as the Toyota Land Cruiser 250 MHV, Jeep Wagoneer ICE, Acura ZDX EV, Jeep Wrangler PHV, Kia EV9 EV, and several premium ICE SUVs. However, it still ranks above several electrified or hybrid SUV launches, including the Lexus RZ EV, Cadillac Lyriq EV, Lexus TX HV, BMW X5 PHV, Hyundai Ioniq 9 EV, Volvo XC40 EV, Acura MDX HV, Ford Explorer HV, Lexus NX PHV, and Volvo C40 Recharge EV. This suggests that the model views the Scout Traveler as competitive within the electrified SUV space, but not strong enough to match the initial launch volume of the most established or high demand SUV nameplates.

One important interpretation is that historic nameplate strength appears to have a large influence on first year launch sales. Vehicles tied to well known SUV identities, such as the Toyota Land Cruiser and Jeep Wrangler generally perform well in the benchmark group. This is consistent with the feature importance results, where 'Maker/Brand' and launch timing variables carried substantial weight. The results also suggest that traditional ICE SUVs have historically performed better at launch than many hybrid and EV SUVs in similar price brackets. While the Scout Traveler offers a differentiated EREV powertrain and strong range proposition, the model may be conservative because Scout is effectively a revived or new market entrant without recent U.S. sales history. As a result, the forecast should be interpreted as a cautious first year estimate rather than a ceiling on demand.

Conclusion

The final modeling strategy focused on the first twelve months after launch and excluded lag based sales features, making the framework more appropriate for a hypothetical vehicle with no prior sales record. The Scout Traveler EREV forecast suggests that the vehicle could achieve meaningful first year demand, but that its performance would likely depend heavily on market positioning, brand perception, and the strength of its range extender value proposition. Compared with similar SUV benchmarks, the Traveler forecast falls below highly established SUV nameplates such as the Toyota Land Cruiser and Jeep Wrangler, but remains competitive against several electrified SUV launches. This supports one of the main findings of the study: early launch sales are influenced not only by technical specifications, but also by nameplate familiarity, brand strength, seasonality, pricing, and segment positioning. The Scout Traveler's projected range above 500 miles, SUV body style, strong performance, and Volkswagen backed brand proxy helped support its forecast.

Overall, the results indicate that the Scout Traveler EREV should not be interpreted as a low volume niche entrant. Instead, the forecast places it as a competitive electrified SUV with potential to perform well within its price and powertrain class, especially if consumers respond positively to the EREV architecture as a solution to range anxiety, towing concerns, and charging limitations. However, the forecast should be treated as directional rather than definitive. The model cannot fully capture future marketing strategy, production constraints, dealer availability, consumer awareness, macroeconomic conditions, or the uniqueness of Scout's revived brand identity. Future work could improve the analysis by incorporating reservation data, regional demand trends, charging infrastructure variables, consumer sentiment, advertising intensity, and separate modeling for SUV and pickup body styles. Even with these limitations, the study demonstrates that historical launch data and vehicle attributes can provide a useful foundation for estimating the early market potential of new electrified vehicle concepts.

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